THE DERIVED LIMIT OF THE QUASI-COHERENT APPROXIMATIONS LIES IN BOUNDED ABOVE QUASI-COHERENT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0DIC (sources/stacks-flat/pinned-original-source.tex line 8406).

Allowed standard prerequisites (editorial).

- Existence of the coherator as a right adjoint with bounded cohomological dimension. Derived Categories of Schemes, Lemmas labelled perfect-lemma-better-coherator (Stacks tag 0CR0) and perfect-lemma-boundedness-better-coherator (Stacks tag 0CSA), invoked at proof lines 8417 and 8424-8426. Supply the existence of DQ_X and its commutation with derived limits, and the bounded cohomological dimension that reduces the claim to R-lim K_n in D^-(O_X).
- The exact sequence relating the derived limit of cohomology to the cohomology of the derived limit, and the comparison of cohomology for pseudo-coherent complexes on an affine. Cohomology, Lemma labelled cohomology-lemma-RGamma-commutes-with-Rlim (Stacks tag 0D60) at proof lines 8429-8435, and Derived Categories of Schemes, Lemmas labelled perfect-lemma-pseudo-coherent (Stacks tag 08E5) and perfect-lemma-affine-compare-bounded (Stacks tag 06Z0) at proof lines 8436-8441. Supply the reduction of the claim to the boundedness of the K_n independent of n.
- The situation of the existence-of-hulls setup and Nakayama for a finite presentation module along a thickening. Same chapter, Situation labelled situation-existence-derived (source line 8379) invoked in the statement, Cohomology, Lemma labelled cohomology-lemma-finite-cohomology (Stacks tag 08DN) at proof line 8451 and Algebra, Lemma labelled algebra-lemma-NAK (Stacks tag 00DV) at proof lines 8454-8456. Supply the tower X_n, the pseudo-coherence of K_n and the finite presentability that makes Nakayama applicable; the comparison of the degrees a_n is carried out in the counted span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 0DIB: sources/stacks-flat/pinned-original-source.tex, source lines 8379–8379. Same-chapter pointer into the author environment situation at source line 8379; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CR0: sources/stacks-chow/tags.txt, source lines 15980–15980. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8418 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CSA: sources/stacks-chow/tags.txt, source lines 16025–16025. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8425 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0D60: sources/stacks-chow/tags.txt, source lines 16505–16505. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8435 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E5: sources/stacks-chow/tags.txt, source lines 10661–10661. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8438 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8441 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DN: sources/stacks-chow/tags.txt, source lines 10643–10643. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8451 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8455 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0DIB consider*

$$K = R\lim_{D_{QCoh}(\mathcal{O}_X)}(K_n) = DQ_X(R\lim_{D(\mathcal{O}_X)} K_n)$$

Then K is in $D^-_{QCoh}(\mathcal{O}_X)$.

Proof. The functor DQ_X exists because X is quasi-compact and quasi-separated, see Derived Categories of Schemes, Lemma 0CR0. Since DQ_X is a right adjoint it commutes with products and therefore with derived limits. Hence the equality in the statement of the lemma.

By Derived Categories of Schemes, Lemma 0CSA the functor DQ_X has bounded cohomological dimension. Hence it suffices to show that $R\lim K_n \in D^-(\mathcal{O}_X)$. To see this, let $U \subset X$ be an affine open. Then there is a canonical exact sequence

$$0 \to R^1 \lim H^{m-1}(U, K_n) \to H^m(U, R\lim K_n) \to \lim H^m(U, K_n) \to 0$$

by Cohomology, Lemma 0D60. Since U is affine and K_n is pseudo-coherent (and hence has quasi-coherent cohomology sheaves by Derived Categories of Schemes, Lemma 08E5) we see that $H^m(U, K_n) = H^m(K_n)(U)$ by Derived Categories of Schemes, Lemma 06Z0. Thus we conclude that it suffices to show that K_n is bounded above independent of n.

Since K_n is pseudo-coherent we have $K_n \in D^-(\mathcal{O}_{X_n})$. Suppose that a_n is maximal such that $H^{a_n}(K_n)$ is nonzero. Of course $a_1 \leq a_2 \leq a_3 \leq \ldots$. Note that $H^{a_n}(K_n)$ is an $\mathcal{O}_{X_n}$-module of finite presentation (Cohomology, Lemma 08DN). We have $H^{a_n}(K_{n-1}) = H^{a_n}(K_n) \otimes_{\mathcal{O}_{X_n}} \mathcal{O}_{X_{n-1}}$. Since $X_{n-1} \to X_n$ is a thickening, it follows from Nakayama's lemma (Algebra, Lemma 00DV) that if $H^{a_n}(K_n) \otimes_{\mathcal{O}_{X_n}} \mathcal{O}_{X_{n-1}}$ is zero, then $H^{a_n}(K_n)$ is zero too. Thus $a_n = a_{n-1}$ for all n and we conclude. $\square$